

Warning Override Workflow

This document describes how curators manage validation warnings that are expected or acceptable and should not block upload.

What warnings mean

When SurveyValidator runs, it flags data conditions that may indicate errors. Warnings are conditions that are suspicious but not definitively wrong — for example, a latitude that falls slightly outside the typical survey area, or a year that predates the standard survey era.

By default, warnings block upload just like errors. This is intentional: every warning should be reviewed by a human before data goes into the database.

The override workflow

1. Run validation. Process surveys with `BatchUploader.uploadFromFolder()`. Surveys with new warnings will be moved to failed/ and logged to the run's `_errors_<timestamp>.log` (see “Reviewing the run log” below for its exact location).
2. Review each warning. Open that run's `_errors_<timestamp>.log` and read the warning messages. For each warning, decide:
 - Fix the data. Correct the source CSV and re-run. This is the preferred path.
 - Acknowledge the warning. If the data is actually correct and the warning is a false positive, add an override entry to `config/overrides/<batch>_overrides.csv`.
3. Add an override entry. Open `config/overrides/<batch>_overrides.csv` in a text editor and add one row per acknowledged warning:

```
fileid,eventno,field,rule_id,acknowledged_by,acknowledged_date,reason
f098027,42,LAT_DD,coordinate_rules.outside_survey_lat,RS,2026-06-25,Opportunistic sighting near shore
```

Required fields: | Column | What to enter | | _____ | _____

_____ || fileid | The value of the FILEID data column in the survey file (e.g., f098027). This must match the column value exactly — it is not the CSV filename. For example, a file named f098027_v2.csv that contains FILEID=f098027

in every row requires `fileid=f098027` in the override. || `eventno` | The EVENTNO of the specific event row || `field` | The field that was flagged (e.g., `LAT_DD`) || `rule_id` | The rule identifier from the warning (e.g., `coordinate_rules.outside_survey_lat`) || `acknowledged_by` | Your name or initials || `acknowledged_date` | Today's date in YYYY-MM-DD format || `reason` | Brief explanation (optional but recommended) |

4. Re-run the upload. Move the survey CSV back to pending/ and run the uploader again. Surveys whose warnings all match entries in `config/overrides/<batch>_overrides.csv` will now upload successfully. Surveys with any unacknowledged warnings will still fail.
5. Commit **`config/overrides/<batch>_overrides.csv`**. The override file is version-controlled. After adding entries, commit it so other curators and future runs share the same acknowledged state.

Finding the rule_id for a warning

The `rule_id` appears in the validation output and in `_errors_<timestamp>.log`. It takes the form `<rule_file>.<check_name>`, for example:

rule_id	What it checks
<code>coordinate_rules.outside_survey_latitude</code>	Latitude outside typical survey area
<code>coordinate_rules.outside_survey_longitude</code>	Longitude outside typical survey area
<code>datetime_rules.year_too_old</code>	Year before warning threshold (default: 1980)
<code>environmental_rules.visibility_too_high</code>	Visibility above maximum
<code>environmental_rules.surftemp_too_low</code>	Surface temperature below minimum
<code>environmental_rules.surftemp_too_high</code>	Surface temperature above maximum
<code>species_rules.taxcode_not_in_table</code>	TAXCODE not found in lookup table
<code>species_rules.number_zero_for_sighting</code>	NUMBER=0 for a sighting record
<code>species_rules.number_unusual</code>	Group size exceeds the SPECCODE/TAXCODE/global threshold cascade (see docs/validation_rules_guide.md)
<code>species_rules.numcalf_unusual</code>	Calf count exceeds the SPECCODE/TAXCODE/global threshold cascade
<code>species_rules.numcalf_non_mammal</code>	Calf count for non-mammal species

rule_id	What it checks
species_rules.numcalf_exceeds_half	Calf count more than half the group — can be suppressed globally by setting validation.species.allow_numcalf_exceeds_half = true in a batch config (already set in migration.m)
species_rules.right_whale_large	Right whale group unusually large
species_rules.right_whale_high_calf	Right whale calf count unusually high
behavioral_rules.calf_behavior_no_calf	Calf associated behavior without calf present
behavioral_rules.taxcode_behavior_not_valid	Behavior not valid for taxcode
behavioral_rules.species_behavior_not_typical	Behavior not typical for species

How to remove an incorrect override

If you acknowledged a warning by mistake, delete the corresponding row from config/overrides/<batch>_overrides.csv and commit the change. The next validation run will treat that warning as new again.

Override matching

Override matching runs in two passes, in order:

1. Per-row pass — any override where all four fields match exactly:
 - fileid — exact string match
 - eventno — exact integer match
 - field — exact string match
 - rule_id — exact string match
2. Per-survey pass — any override where eventno is blank AND fileid, field, and rule_id all match. The warning's specific eventno is ignored.

A warning is acknowledged when either pass finds a match. If both passes would match the same warning (e.g., you wrote both a per-row entry and a per-survey entry for the same rule), the per-row pass wins and the warning is counted once.

Per-survey overrides

When a survey produces many warnings of the same type — for example, a historic survey file where every sighting row has a year before 1980 — it is impractical to write a separate override entry per row. A per-survey override acknowledges all warnings of a given (`field`, `rule_id`) combination across the entire survey with a single CSV row.

To write a per-survey override, leave the `eventno` column empty:

```
fileid,eventno,field,rule_id,acknowledged_by,acknowledged_date,reason
p905169G,,BEHAV,behavioral_rules.calf_behavior_no_calf,RS,2026-
06-25,All 93 calf-behavior events reviewed; survey predates current coding convention
```

The `reason` field is effectively required for per-survey overrides — a blanket acknowledgement with no documented rationale is hard to audit.

Trade-off: a per-survey override is convenient but silently absorbs all future warnings of that type in that survey, including any that arise from later edits to the source file. Before issuing one, review at least a representative sample of the affected warnings and record your reasoning in the `reason` field.

A future version may support pattern-based overrides spanning multiple surveys. For now, write one entry per survey.

When to use `AllowWarnings = true`

The `AllowWarnings = true` option in `BatchUploader.uploadFromFolder()` bypasses the entire warning gate and uploads surveys regardless of warnings. This is an emergency escape valve for situations where you need data in the database immediately and cannot review warnings first.

Use it rarely and with a documented reason. Warnings exist for a reason. If you use `AllowWarnings = true`, make sure you revisit the warnings and either fix the data or add proper override entries afterward.

Reviewing the run log

After each batch run, two files record what happened. They land one level above the `pending//processed//failed/` folders — e.g. if surveys live in `data/legacy/surveys/`, the logs land in `data/legacy/` (`BatchUploader.initializeErrorLog`), not inside `failed/` itself:

- `_errors_<timestamp>.log` — Human-readable log, one entry per failed survey for that run. Each run gets its own timestamped file by default (`config.pipeline.logging.use_datetime_filenames`)

= true); setting that to false accumulates all runs into a single non-timestamped `_errors.log` instead.

- `_run_summary_<timestamp>.csv` — Machine-readable CSV with one row per survey per run: fileid, status, error count, new warning count, acknowledged warning count (split into per-row/per-survey), notes.

Use `_run_summary_<timestamp>.csv` to track progress across migration runs and confirm that override additions are having the expected effect.